

Substack piece: final figure lineup

The fine print of event contracts predicts disputes

Locked — July 14, 2026

Narrative order: a disaster you could have seen in the text (Fig. 1); the model works in the real world, unweighted (Fig. 2), and its probabilities are honest (Fig. 3); how the AI reads a contract (Fig. 4) and which flaws carry the signal (Fig. 5); the letter grades (Fig. 6). Every statistic is out-of-sample: markets are scored by a model that never saw their series or market family.

“Will Zelenskyy wear a suit before July?”

Polymarket, 2025 — \$240 million traded

What the rules pinned down:

- ✓ Exact date window (May 22 – June 30, 2025 ET)
- ✓ Must be photographed or videotaped
- ✓ AI-generated or edited images excluded

What the rules never said:

- ✗ What counts as a “suit” — the entire question
- ✗ Who decides: only “a consensus of credible reporting”

Risk grade (from the text alone): **BB — below investment grade**

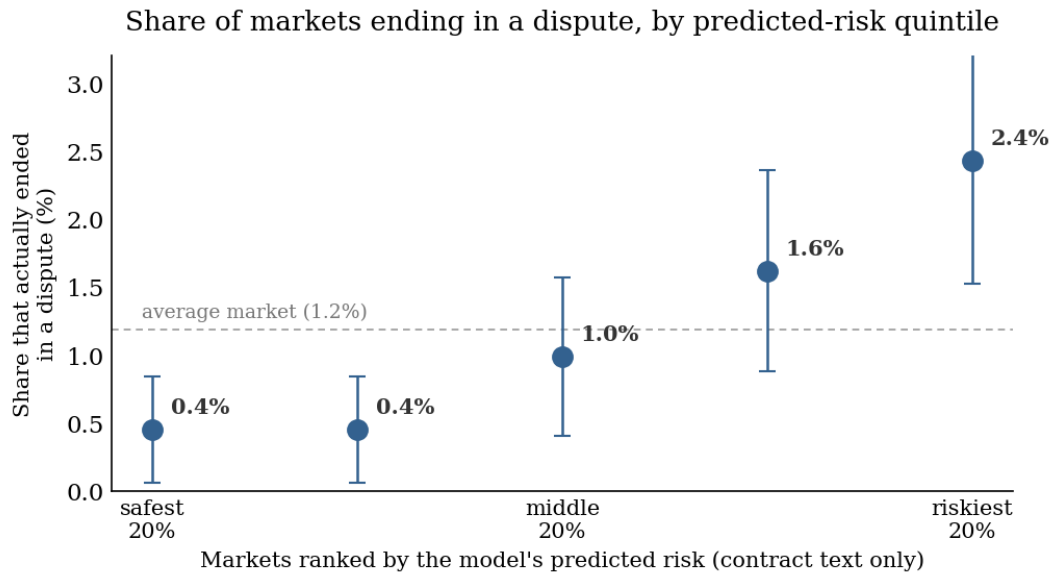
What happened: **disputed 5 times — most in our data**



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Data: Polymarket & Kalshi (Hall & Paschal)

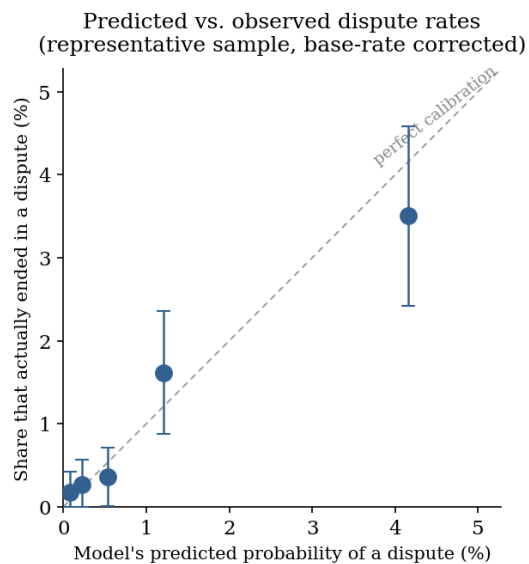
Figure 1: The suit market’s rules, annotated. The actual rules of Polymarket’s \$240M Zelenskyy-suit market: precise about the wrong things (dates, deepfakes), silent on the word the market was named after and on who decides. From the text alone it grades BB — below investment grade; scored out-of-sample (by a model that never saw this market or its siblings) it lands in the riskiest tenth of Polymarket. In fact it was disputed five times, the most of any market in the data.



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Figure 2: Real-world dispute rates by predicted-risk quintile. All 5,567 markets in the representative random sample (both platforms), ranked by the model's predicted risk from contract text alone and split into quintiles; the y-axis is the raw share that actually ended in a dispute (95% binomial CIs). The riskiest fifth disputes at $\sim 6\times$ the rate of the safest fifth. Companion fact for the text: on Polymarket — where disputes are organic third-party challenges, the purest prediction test — 6 of the 9 real-world disputes landed in the model's riskiest 20% of markets (out-of-sample AUC 0.78).



freesystems.substack.com Data: Polymarket & Kalshi (Hall & Paschal)

Figure 3: Calibration of the corrected predicted probabilities. The same out-of-sample scores converted to real-world probabilities (per-platform base-rate correction), binned, against the share of markets that actually disputed; the dashed line is perfect calibration. When the model says ~4%, about 3.5% of those markets blow up; every bin's CI crosses the line. The model doesn't just rank markets correctly — its numbers mean what they say.

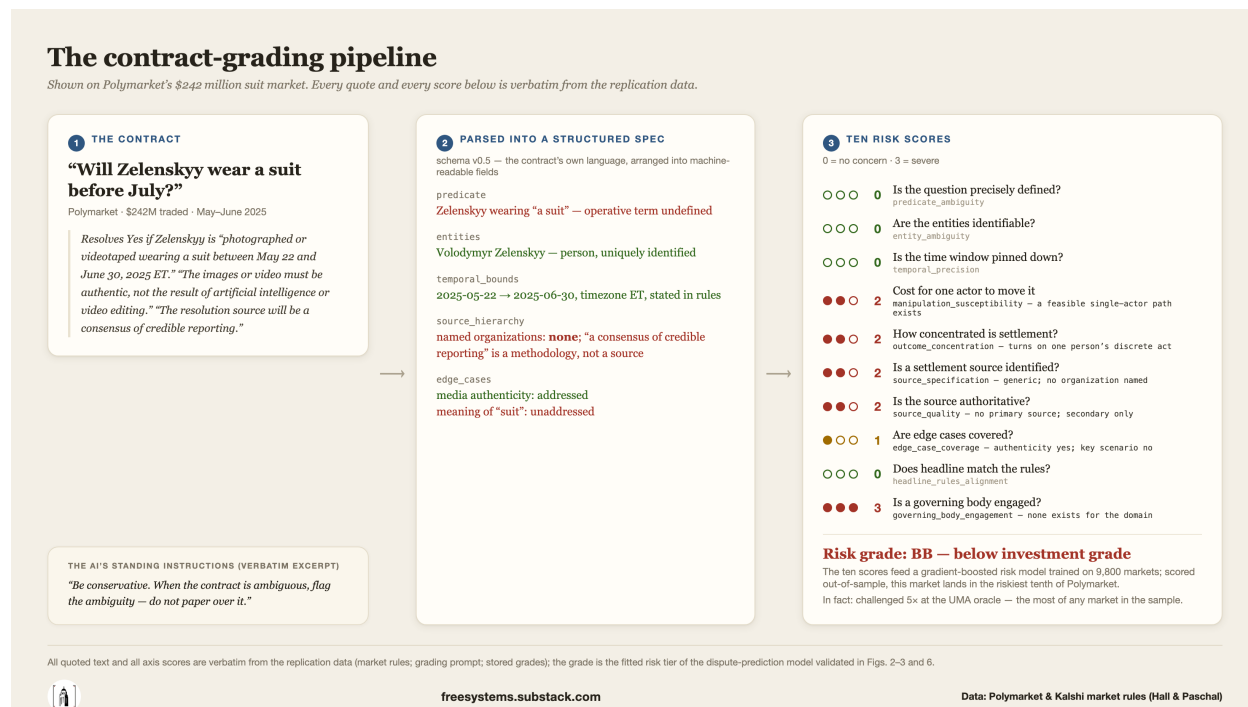


Figure 4: The grading pipeline, applied to one contract. The pipeline on the suit market itself. Provenance: the rules text and standing instructions are verbatim quotes (from the market’s oracle record and the actual grading prompt); the ten axis scores and their schema names are the stored values in the replication data; step 2 arranges the contract’s own language into the real schema fields (the per-market parse itself is not part of the public data). The ten scores are the inputs to the gradient-boosted dispute-risk model validated in Figs. 2–3; the grade (BB, below investment grade) is that model’s fitted risk tier from Fig. 6, and out-of-sample this market’s predicted risk sits in the riskiest tenth of Polymarket. One miss, disclosed: the AI scored the question itself as precisely defined — “suit” proved otherwise — yet the missing referee (no named source, no settling authority, a one-person outcome) pushed the market below investment grade from the text alone.

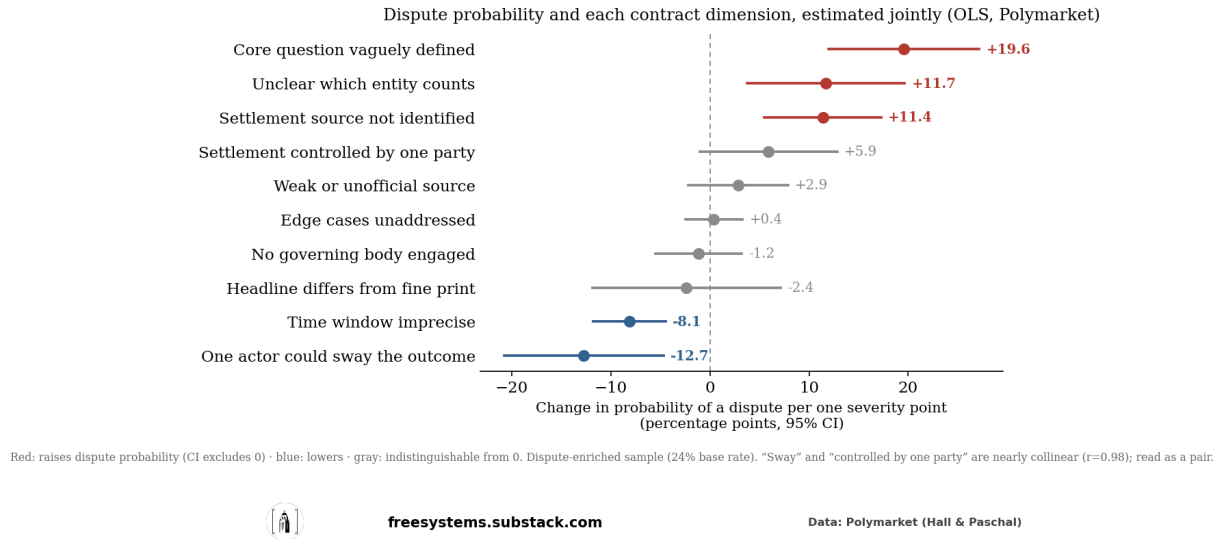


Figure 5: Dispute probability and each contract dimension, jointly estimated (OLS). One linear probability regression of the dispute indicator on all ten 0–3 severity scores at once; Polymarket organic specification disputes vs. clean markets ($n = 3,643$; 877 disputes). Points: change in dispute probability per one severity point, in percentage points; whiskers: 95% CIs clustered by market family. Dispute-enriched sample (24% base rate), so magnitudes scale with that base; the comparisons across dimensions are the point. Vaguely defined core questions (+19.6pp), ambiguous entities (+11.7), and unidentified settlement sources (+11.4) raise dispute probability holding the other dimensions fixed; manipulability (−12.7) and imprecise time windows (−8.1) lower it (manipulable markets tend to resolve unambiguously); the remaining dimensions are indistinguishable from zero. The two structural scores are nearly collinear ($r = 0.98$) and should be read as a pair. (Kalshi’s exchange flags behave differently and are analyzed separately in the paper.)

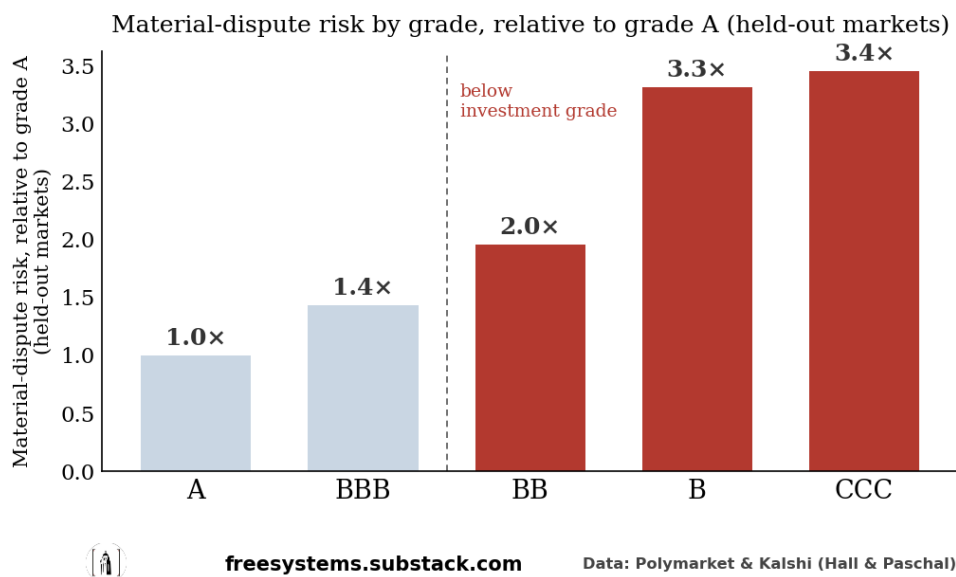


Figure 6: Relative material-dispute risk by grade. The deployed (confirmed-trained) grade ladder, evaluated on material disputes — genuine specification problems, a broader and better-measured outcome than confirmed blow-ups (571 held-out events vs. 85). Evaluation set: clean controls plus material disputes, none used to fit the score or the cutpoints; both platforms. Risk relative to grade A: 1.0×, 1.4×, 2.0×, 3.3×, 3.4× — every grade steps up, and the investment-grade line marks the jump past 2×. Rates are case-control inflated, so only the ratios are meaningful; real-world dispute rates are $\sim 0.3\text{--}2\%$. (At the rarer confirmed-blow-up tier the below-line jump is an order of magnitude — $5\times/23\times/26\times$ — but A and BBB are statistically indistinguishable there on 21 held-out events; see the paper.)